

SymPy Bug Report

1 Bug 44: continuous_domain Misses Positive Periodic Intervals

1.1 Minimal Reproducer

```
from sympy import S, pi, sin, sqrt, symbols
from sympy.calculus.util import continuous_domain

x = symbols("x")
expr = sqrt(sin(x))
dom = continuous_domain(expr, x, S.Reals)

print("domain =", dom)
print("contains 5*pi/2 =", dom.contains(5*pi/2))
print("value at 5*pi/2 =", expr.subs(x, 5*pi/2))
```

1.2 Observed Behavior

```
domain = Interval(0, pi)
contains 5*pi/2 = False
value at 5*pi/2 = 1
```

SymPy reports only the first nonnegative sine interval, $[0, \pi]$, as the continuous real domain of $\sqrt{\sin(x)}$. This excludes $5\pi/2$, even though direct substitution gives $\sqrt{\sin(5\pi/2)} = 1$, so the expression is real-valued and locally continuous there.

1.3 Expected Behavior

The real continuous domain should include every interval on which $\sin(x) \geq 0$:

$$\bigcup_{n \in \mathbb{Z}} [2n\pi, (2n+1)\pi].$$

In particular, $5\pi/2 = \pi/2 + 2\pi$ must be included.

1.4 Mathematical Explanation

For real x , the principal square root $\sqrt{\sin(x)}$ is real-valued exactly where its radicand is nonnegative. Thus its real domain is determined by

$$\sin(x) \geq 0.$$

Since sine has period 2π , the nonnegative portions are

$$[2n\pi, (2n+1)\pi] \quad (n \in \mathbb{Z}).$$

The point $5\pi/2$ lies in the interval $[2\pi, 3\pi]$, and

$$\sin(5\pi/2) = 1, \quad \sqrt{\sin(5\pi/2)} = 1.$$

Therefore the returned set is not an alternate but equivalent representation: it omits whole translated intervals where the expression is real-valued and continuous.

1.5 Source-Level Diagnosis

The diagnosis identifies the relevant implementation points as `sympy/calculus/util.py`, in the continuous-domain routine, and `sympy/solvers/inequalities.py`, in the univariate inequality solver. For the outer square root, the continuous-domain routine iterates over `Pow` atoms and imposes a nonnegative-base condition for non-odd rational powers. In this case that condition is $\sin(x) \geq 0$, so the call path asks the univariate inequality solver to solve the periodic inequality and then intersects the returned set with $\mathbb{R}$.

The root-cause note locates the problematic behavior in the solver’s periodic branch. The solver detects period 2π for $\sin(x)$, replaces the unbounded real solving domain with the single half-open interval $[0, 2\pi)$, solves there, and returns the one-period answer directly. There is no later lifting step that converts the fundamental-domain piece $[0, \pi]$ into all integer translates.

```
# distilled from the diagnosed behavior
if domain is S.Reals and period is not None:
    domain = Interval(0, period, False, True)

# for sin(x) >= 0 this yields only the fundamental-domain piece
# Interval(0, pi), which continuous_domain then uses as the full constraint
```

This source-level behavior explains the mathematical failure: the square-root domain constraint is correct, but the inequality solver supplies only the principal-period slice of that constraint. A plausible fix direction supported by the diagnosis is for periodic inequalities over unbounded real domains to return a periodic set, such as an `ImageSet` or periodic union, or to decline with a condition set rather than returning only the fundamental-domain slice. Then `continuous_domain` would receive the full radical constraint.

1.6 Additional Failing Instances

The independent verification checks the family

$$\sqrt{\sin(kx)}, \quad k = 1, 2, 3, 4, 5, 6,$$

using the witness $x = 5\pi/(2k)$. For every positive integer k in this list,

$$\sin\left(k \cdot \frac{5\pi}{2k}\right) = \sin(5\pi/2) = 1,$$

so the witness belongs to a translated positive interval of the real domain. SymPy instead returns only the first interval $[0, \pi/k]$ and excludes the witness.

Input / Expression	SymPy behavior	Expected behavior
$\sqrt{\sin(x)}$, witness $5\pi/2$	$[0, \pi]$; witness excluded	Include witness; value 1
$\sqrt{\sin(2x)}$, witness $5\pi/4$	$[0, \pi/2]$; witness excluded	Include witness; value 1
$\sqrt{\sin(3x)}$, witness $5\pi/6$	$[0, \pi/3]$; witness excluded	Include witness; value 1
$\sqrt{\sin(4x)}$, witness $5\pi/8$	$[0, \pi/4]$; witness excluded	Include witness; value 1
$\sqrt{\sin(5x)}$, witness $\pi/2$	$[0, \pi/5]$; witness excluded	Include witness; value 1
$\sqrt{\sin(6x)}$, witness $5\pi/12$	$[0, \pi/6]$; witness excluded	Include witness; value 1

The expected behavior is the same across the family because the radical is real-valued exactly where $\sin(kx) \geq 0$, and this condition repeats on translated intervals of length π/k separated by the period $2\pi/k$. The witnesses are all in translated positive intervals, not in a branch where the square root is undefined or discontinuous.

1.7 Related Issues Assessment

The bug-finding harness performed a best-effort deduplication check. The underlying root cause — a periodic inequality solved on one period and not lifted to all real periods by `solve_univariate_inequality` — is a known, publicly reported defect: open issue #27573 (<https://github.com/sympy/sympy/issues/27573>, the `solveset(sin(x) > 0)` returning `Interval(0, pi)` symptom), closed issue #9721 (<https://github.com/sympy/sympy/issues/9721>), and proposed fixes #27720 (<https://github.com/sympy/sympy/pull/27720>) and #27745 (<https://github.com/sympy/sympy/pull/27745>). No public report was found for the exact reproducer `continuous_domain(sqrt(sin(x)), x, S.Reals)`, for the returned single interval `Interval(0, pi)`, or for the missed $5\pi/2$ witness, so this `continuous_domain +  $\sqrt{\cdot}$` composition appears unreported. The deduplication verdict is therefore `family_known_specific_new`: the failure family is known, but the case remains individually actionable because it gives a concise, reproducible `continuous_domain` failure with a direct regression test.

1.8 Proposed SymPy Regression Test

```
import pytest
from sympy import S, pi, sin, sqrt, symbols
from sympy.calculus.util import continuous_domain

@pytest.mark.parametrize("k", [1, 2, 3, 4, 5, 6])
def test_continuous_domain_sqrt_sin_periodic_intervals(k):
    x = symbols("x")
    dom = continuous_domain(sqrt(sin(k*x)), x, S.Reals)
    assert dom.contains(5*pi/(2*k)) is S.true
```